Evolution of Log Analysis Architectures

Going down memory lane, it's easy to see how changing needs have helped evolve log analysers since the mid-2000s. What worked then, will not work today. But the journey is to be cherished.

It was the late nineties, and I had about 3-4 years of development experience by then. I had just transitioned from the GSM field to the world of network management systems (NMS) after joining a multinational company in Bengaluru. Although my master's degree was in computer networks, NMS was entirely new to me. I was familiar with hubs, routers, switches, IP addresses, RFCs, and protocols, but only in theory. This was the first time I had hands-on experience with NMS.

From a technological standpoint, I was quite proficient. I coded in C++ and worked on HP-UX, a variant of UNIX. I had experience with large-scale, mission-critical systems. In a way, I was filled with youthful confidence when I began working with NMS.

In my new role, I was assigned the task of enhancing a log analysis tool as part of a larger NMS. NMS are structured around the FCAPS model, which stands for fault management, configuration management, accounting, performance, and security. This model is defined by ISO. The log analysis tool our team was developing fell under the fault management category. Figure 1 captures the arrangement.

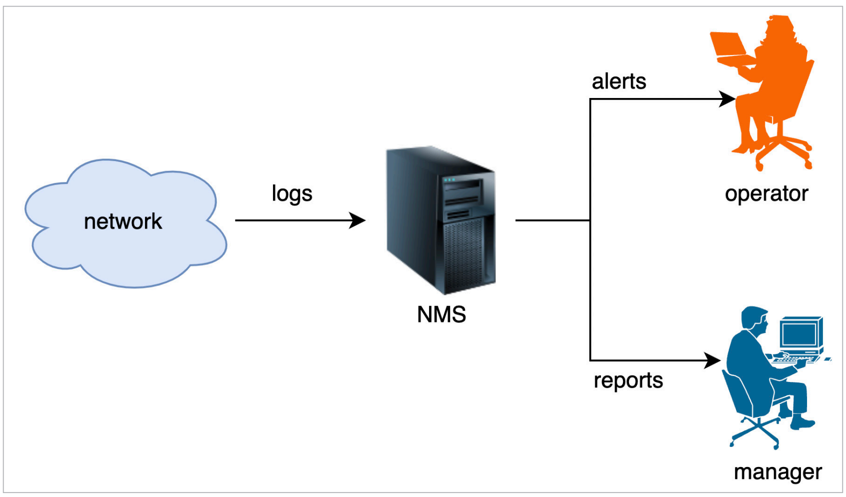

Figure 1: Fault management system